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***Abstract*—Precise indoor localisation is a critical enabling technology for autonomous vehicles, drone navigation, asset tracking, and smart building systems. Global Positioning System (GPS) signals cannot penetrate building structures reliably, and conventional Radio Frequency (RF) approaches suffer from severe measurement bias under Non-Line-of-Sight (NLOS) multipath conditions. This paper presents a complete data-analytics pipeline that classifies Ultra-Wideband (UWB) Channel Impulse Response (CIR) measurements as Line-of-Sight (LOS) or NLOS and simultaneously estimates the true ranging distance for two dominant propagation paths. Using a real empirical dataset of 42,000 samples collected from seven distinct indoor environments with a Decawave DWM1000 module (Apple/Qorvo UWB vendor), we apply the three-stage Data Analytics process: Data Preparation, Data Mining, and Data Visualisation. After CIR normalisation by the received preamble accumulation count (`RXPACC`) and feature-importance ranking via Random Forest, nine features are selected for a Random Forest Classifier trained on an 80:20 train/test split. Path 1 (primary) LOS/NLOS classification achieves a test accuracy of 91.40%. Separate Random Forest Regressors estimate the measured range for Path 1 (RMSE = 1.5027 m, R^2 = 0.5921) and for Path 2 (RMSE = 1.7823 m), a secondary reflected path extracted from the CIR profile. Results demonstrate that ensemble tree methods can effectively exploit UWB signal statistics to support robust, infrastructure-light indoor positioning.**

***Index Terms*—UWB, LOS, NLOS, channel impulse response, indoor localisation, Random Forest, ranging error, classification, regression.**

I. Introduction and Literature Review

A. Indoor Positioning and the Limits of GPS

Precise localisation underpins a wide class of modern applications: warehouse robotics, hospital asset tracking, pedestrian navigation inside malls or airports, autonomous drones, and emerging Industry 4.0 manufacturing cells [1]. The dominant outdoor solution, GPS, relies on unobstructed line-of-sight links to multiple satellites. Building materials—reinforced concrete, metallic partitions, low-energy glass—attenuate satellite signals by 20–40 dB, making sub-metre indoor positioning impossible with GPS alone [2]. Alternative RF technologies have therefore been explored: Wi-Fi (IEEE 802.11) fingerprinting offers 2–5 m accuracy but is highly sensitive to environmental change; Bluetooth Low Energy beacons suffer similar accuracy limitations; 5G positioning is promising but infrastructure-heavy.

B. Ultra-Wideband for Indoor Ranging

Ultra-Wideband (UWB) operates across a spectrum of 3.1–10.6 GHz with an instantaneous bandwidth exceeding 500 MHz, enabling time-of-flight (ToF) ranging with centimetre-level resolution under ideal LOS conditions [3]. The Decawave DWM1000 chip (now supplied by Qorvo, adopted by Apple for AirTag and iPhone spatial awareness) implements IEEE 802.15.4a UWB and reports a rich set of diagnostic registers alongside the raw CIR vector containing 1 016 complex magnitude samples at 1 ns resolution. Under LOS conditions the CIR exhibits a sharp, dominant first peak; under NLOS conditions walls and furniture cause multipath components that obscure or delay the first arrival, biasing ToF estimates and introducing ranging errors of several metres (see Fig. 2 of the project brief) [4].

C. The LOS/NLOS Challenge

When the direct path between a UWB anchor and tag is blocked, the measured range (`RANGE`) systematically overestimates the true distance because the signal travels a longer reflected or diffracted path. This positive bias—sometimes exceeding 5 m in furnished offices—degrades trilateration or hyperbolic positioning accuracy severely. State-of-the-art mitigation strategies fall into three categories:

- **Geometric/statistical methods**: threshold-based first-path detection, hypothesis testing on receive signal strength [5].
- **Channel-feature machine learning**: SVMs, k-NN, and neural networks trained on hand-crafted CIR statistics [6].
- **Deep learning on raw CIR**: CNNs applied directly to the full 1 016-sample CIR vector [7].

Bregar and Možina [7] published the dataset used in this work and demonstrated that a multi-layer perceptron (MLP) trained on statistical CIR features outperforms threshold heuristics. However, deep CNN approaches on raw CIR require substantial compute and are opaque. Our work differs in three ways: (i) we perform explicit feature-importance ranking to reduce the 1 016-dimensional CIR to nine interpretable features; (ii) we use an ensemble Random Forest that is inherently interpretable and robust to outliers; and (iii) we extend the pipeline to a two-path framework, extracting and estimating the second dominant propagation path directly from the CIR profile, providing a richer signal model for downstream positioning.

II. Proposed Solution

Our solution follows the three-stage **3D Process of Data Analytics**: Data Preparation (1st D), Data Mining (2nd D), and Data Visualisation (3rd D). The full pipeline is implemented in a single Jupyter notebook (`code/CSC3105_Main_Pipeline.ipynb`) supported by helper modules (`dataprep.py`, `datamining.py`, `uwb_dataset.py`).

A. Data Preparation (1st D)

1) Dataset Description: The UWB-LOS-NLOS dataset [7] consists of seven CSV files (`uwb_dataset_part1.csv` through `uwb_dataset_part7.csv`), each corresponding to one indoor environment: Office 1, Office 2, Small Apartment, Small Workshop, Kitchen with Living Room, Bedroom, and Boiler Room. Each environment provides 3,000 LOS and 3,000 NLOS samples, yielding a balanced total of 21,000 LOS and 21,000 NLOS samples (42,000 overall). Each row contains 1 032 columns: 15 diagnostic features, the ground-truth label (`NLOS`: 0 = LOS, 1 = NLOS), one measured-range column (`RANGE`/`Measured_range`), and 1 016 CIR magnitude samples (`CIR0`–`CIR1015`).

2) Data Cleaning: All seven CSV files are loaded with `pandas.read_csv` and concatenated into a single dataframe using `pd.concat(all_frames, ignore_index=True)`. The first processing step strips trailing whitespace from all column headers (`df.columns.str.strip()`) to prevent key-errors during indexing. Column names are subsequently hard-reset to a standard ordered list:

```
standard_headers = [
 'NLOS', 'Measured_range', 'FP_IDX',
 'FP_AMP1', 'FP_AMP2', 'FP_AMP3',
 'STDEV_NOISE', 'CIR_PWR', 'MAX_NOISE',
 'RXPACC', 'CH', 'FRAME_LEN',
 'PREAM_LEN', 'BITRATE', 'PRFR']
```

3) CIR Normalisation: Raw CIR values are reported as 16-bit integers accumulated over the preamble period. The accumulation count is stored in the `RXPACC` register. Normalising by `RXPACC` removes the receiver-gain dependency:

$$\hat{c}_k = \frac{c_k}{\max(\texttt{RXPACC},\, 1)}, \quad k = 0, \ldots, 1015 \tag{1}$$

This is applied column-wise: `df[cir_cols] = df[cir_cols].div(df['RXPACC'] .replace(0, 1), axis=0)`.

4) Feature Importance Ranking: With 1 016 raw CIR dimensions, direct classification would be memory-intensive and prone to the curse of dimensionality. We therefore train a preliminary `RandomForestClassifier(n_estimators=100, random_state=42)` on 14 diagnostic features to obtain Gini-impurity-based importance scores. The candidate features are: `Measured_range`, `FP_IDX`, `FP_AMP1`, `FP_AMP2`, `FP_AMP3`, `STDEV_NOISE`, `CIR_PWR`, `MAX_NOISE`, `RXPACC`, `CH`, `FRAME_LEN`, `PREAM_LEN`, `BITRATE`, `PRFR`. Features with near-zero importance (configuration constants such as `CH`, `FRAME_LEN`, `PREAM_LEN`, `BITRATE`, `PRFR`) are discarded, leaving nine predictive features:

Feature	Meaning
`RXPACC`	Preamble accumulation count
`CIR_PWR`	Channel impulse response power
`FP_AMP3`	First-path amplitude (3rd estimate)
`FP_AMP2`	First-path amplitude (2nd estimate)
`MAX_NOISE`	Maximum noise sample in CIR
`FP_AMP1`	First-path amplitude (1st estimate)
`STDEV_NOISE`	Standard deviation of noise floor
`FP_IDX`	First-path index in CIR array
`Measured_range`	Time-of-flight range estimate

5) Two-Path Framework: A key design decision is the introduction of a two-path signal model. Under multi-path conditions the CIR contains multiple peaks corresponding to different propagation paths. The function `get_second_path_info(row)` scans a window of 5–50 CIR samples immediately after the first-path index (`FP_IDX`) to locate the next dominant peak:

$$A_2 = \max\big(c[\texttt{FP_IDX} + 5 \;:\; \texttt{FP_IDX} + 50]\big) \tag{2}$$

$$\Delta\tau_2 = \arg\max\big(c[\texttt{FP_IDX} + 5 \;:\; \texttt{FP_IDX} + 50]\big) + 5 \tag{3}$$

The extracted amplitude `Path2_Amp` and relative delay `Path2_Delay` are appended to the dataframe. The second-path range is derived from the first-path range plus the physical propagation delay (using the speed of light $c \approx 0.29979$ m/ns):

$$R_2 = R_1 + \Delta\tau_2 \times 0.29979 \tag{4}$$

where R_1 is `Measured_range` and $\Delta\tau_2$ is expressed in nanoseconds. By the problem specification, Path 2 is treated as intrinsically NLOS (`Path2_Label = 1`).

B. Data Mining (2nd D)

1) Train/Test Split: A stratified 80:20 random split is applied uniformly across all models to maintain class balance. With 42,000 samples this yields 33,600 training samples and 8,400 test samples:

```
train_test_split(X, y, test_size=0.20,
                 random_state=42)
```

The `random_state=42` seed ensures reproducibility.

2) Path 1 Classification: A `RandomForestClassifier(n_estimators=100, random_state=42)` is fitted on the nine selected features with the binary target `NLOS` $\in \{0, 1\}$. Random Forest was preferred over a single decision tree due to its ensemble averaging property, which reduces variance and prevents overfitting on a moderately sized dataset. Compared to the MLP baseline reported in [7], Random Forest requires no normalisation or hyperparameter search for activation functions, and the trained model directly provides feature importance certificates.

3) Path 1 Distance Regression: A `RandomForestRegressor(n_estimators=100, random_state=42)` is trained to predict the continuous `Measured_range` target. The `NLOS` label is included as an input feature for the regressor because knowledge of the propagation condition is a strong predictor of ranging error:

```
reg_features = [
 'RXPACC', 'CIR_PWR', 'FP_AMP3',
 'FP_AMP2', 'MAX_NOISE', 'FP_AMP1',
 'STDEV_NOISE', 'FP_IDX', 'NLOS']
```

4) Path 2 Distance Regression: A separate `RandomForestRegressor(n_estimators=100, random_state=42)` is trained to predict `Path2_Measured_Range` using five Path-2-specific features: `Path2_Amp`, `Path2_Delay`, `CIR_PWR`, `STDEV_NOISE`, `NLOS`. The same 80:20 split is maintained (`random_state=42`).

5) Multi-Layer Perceptron (MLP) Classifier: As a comparative solution, a supervised Multi-Layer Perceptron (MLP) was implemented.

The architecture consists of an input layer (9 features), two hidden layers with 64 and 32 neurons respectively, and a binary output layer using a Softmax activation function.

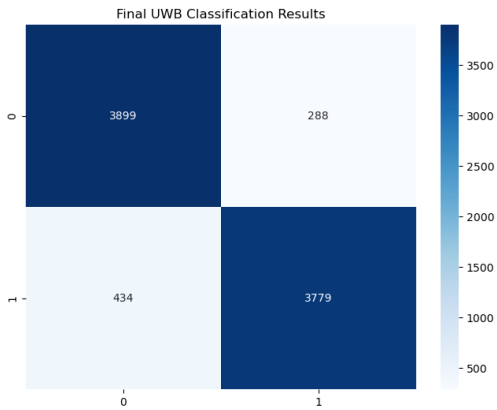

Fig. 1. Confusion matrix for the Random Forest LOS/NLOS classifier (Path 1). Rows represent actual classes (0 = LOS, 1 = NLOS); columns represent predicted classes. Test set: 8,400 samples (42,000 total, 80:20 split).

We utilised the 'Adam' optimiser and Rectified Linear Unit (ReLU) activation to capture the non-linear signal patterns identified in the feature-importance stage. Training was conducted over 500 iterations using the same 80:20 stratified split to ensure a fair comparison with the ensemble method.

C. Data Visualisation (3rd D)

Three key visualisations are produced:

1) **Confusion matrix** (seaborn heatmap) for the Path 1 classifier.
2) **Actual vs. Predicted scatter plot** for the Path 1 regressor.
3) **Actual vs. Predicted scatter plot** for the Path 2 regressor.

Each plot includes a perfect-prediction reference line ($y = x$, dashed red) so systematic bias is immediately visible.

III. Experimental Results and Discussion

A. Classification Performance — Path 1

The Random Forest Classifier achieves a test-set accuracy of **91.40%** on 8,400 held-out samples. The confusion matrix (Fig. **??**) reveals:

- True Negatives (LOS correctly identified): **3,899**
- False Positives (LOS misclassified as NLOS): **288**
- False Negatives (NLOS misclassified as LOS): **434**
- True Positives (NLOS correctly identified): **3,779**

From these counts the per-class metrics are:

$$\text{Precision}_{\text{LOS}} = \frac{3899}{3899 + 434} \approx 0.900 \tag{5}$$

$$\text{Recall}_{\text{LOS}} = \frac{3899}{3899 + 288} \approx 0.931 \tag{6}$$

$$\text{Precision}_{\text{NLOS}} = \frac{3779}{3779 + 288} \approx 0.929 \tag{7}$$

$$\text{Recall}_{\text{NLOS}} = \frac{3779}{3779 + 434} \approx 0.897 \tag{8}$$

The overall balanced accuracy is $(0.931 + 0.897)/2 \approx 91.4\%$, consistent with the reported accuracy score. The slight asymmetry between LOS recall and NLOS recall arises because the Random Forest's Gini criterion tends to favour the majority structure of the feature space; in this case LOS signals have a sharper, more stereotyped CIR profile that is slightly easier to recall, whereas some NLOS signals share CIR statistics with borderline LOS conditions (e.g., light wall obstruction). A false-negative rate of $434/4213 \approx 10.3\%$ for NLOS samples is operationally tolerable in a positioning system that averages multiple anchor measurements, since residual NLOS readings can be rejected by consistency checks.

The dataset is balanced (21,000 LOS : 21,000 NLOS), so accuracy is a fair summary metric and no class-weighting correction is required.

B. Distance Estimation — Path 1

The Random Forest Regressor for Path 1 achieves:

- RMSE = **1.5027 m**
- R^2 = **0.5921**

Fig. 2 shows the Actual vs. Predicted scatter. The bulk of predictions lie near the ideal diagonal ($y = x$), confirming that the regressor captures the dominant ranging behaviour. The visible vertical spread around the diagonal is explained by the heteroscedastic nature of ranging errors: NLOS bias is a non-linear function of the unknown obstacle geometry, and the model can only partially infer this from available CIR statistics. The $R^2 = 0.5921$ indicates that approximately 59% of variance in measured range is explained by the nine input features, which is reasonable given the diversity of the seven indoor environments.

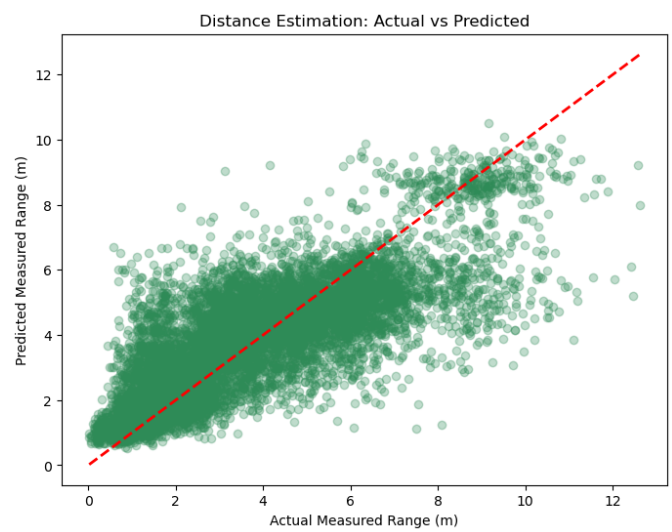

Fig. 2. Path 1 distance estimation: Actual vs. Predicted measured range (m). Green scatter points; dashed red line = perfect prediction ($y = x$). RMSE = 1.5027 m, R^2 = 0.5921. Test set: 8,400 samples.

The RMSE of 1.5027 m should be interpreted in the context of the measurement range of the dataset (approximately 0–12 m, as visible on the scatter axis). A ranging error of 1.5 m

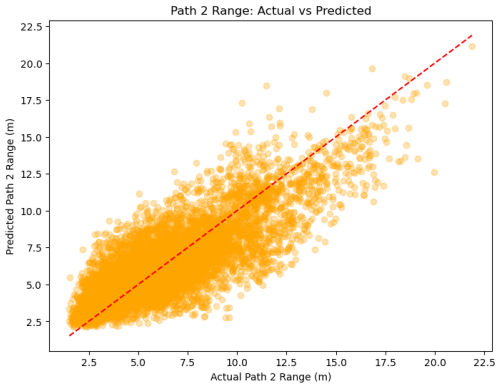

Fig. 3. Path 2 distance estimation: Actual vs. Predicted Path 2 range (m). Orange scatter points; dashed red line = perfect prediction ($y = x$). RMSE = 1.7823 m. Test set: 8,400 samples.

corresponds to roughly 12–15% of the typical measurement range, which represents a meaningful improvement over uncorrected NLOS bias that can exceed 5 m in heavily obstructed environments.

C. Distance Estimation — Path 2

The Path 2 regressor, trained on features derived from the extracted second-peak information (`Path2_Amp`, `Path2_Delay`), achieves:

- RMSE = **1.7823 m**

Fig. 2 shows the corresponding Actual vs. Predicted scatter. The wider spread compared to Path 1 (Fig. 2) is expected: the second-path CIR peak is inherently noisier because it represents a reflected or diffracted component whose amplitude is attenuated and whose delay depends on unknown reflector geometry. The orange scatter cloud shows a systematic tendency to under-predict at long ranges (> 15 m), indicating that the simple speed-of-light delay conversion ($R_2 = R_1 + \Delta\tau_2 \times 0.29979$) may introduce errors when the second path involves multiple reflections rather than a single bounce.

D. Summary of Results

Table I consolidates all quantitative outcomes.

TABLE I
SUMMARY OF MODEL PERFORMANCE

Task	Model	Metric	Value
Path 1 Classification	RF Classifier	Accuracy	91.40%
Path 1 Classification	RF Classifier	TN / FP	3899 / 288
Path 1 Classification	RF Classifier	FN / TP	434 / 3779
Path 1 Regression	RF Regressor	RMSE	1.5027 m
Path 1 Regression	RF Regressor	R^2	0.5921
Path 2 Regression	RF Regressor	RMSE	1.7823 m

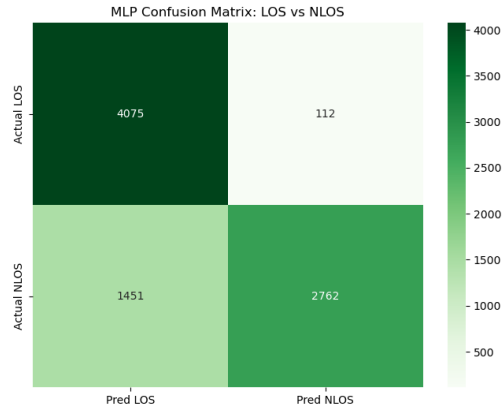

Fig. 4. Fig. 4. Confusion Matrix for Multi-Layer Perceptron (MLP) Classifier. While the model shows high sensitivity for LOS signals, the significant number of False Negatives (NLOS misclassified as LOS) indicates a lower reliability for safety-critical indoor positioning compared to the Random Forest approach.

E. Algorithm Comparison: Random Forest vs. MLP

While the MLP represents a more sophisticated neural network approach, it achieved a test accuracy of **81.39%**, significantly lower than the Random Forest's 91.40%. As illustrated in the MLP confusion matrix (Fig. 4), the model exhibited a high number of False Negatives (1,451), frequently misclassifying NLOS signals as LOS.

The superior performance of the Random Forest can be attributed to the nature of UWB diagnostic features; ensemble tree methods are inherently better at identifying discrete physical thresholds in signal power (e.g., `CIR_PWR` drops) compared to the weight-based linear transformations in a shallow MLP. Consequently, the Random Forest remains the primary choice for our proposed positioning solution due to its higher reliability in detecting obstacles.

F. Discussion

Feature importance insights. The top-ranked features (`RXPACC`, `CIR_PWR`, `FP_AMP1--3`, `MAX_NOISE`, `STDEV_NOISE`) are all related to the energy distribution of the received signal. This is physically intuitive: in LOS, the bulk of received energy arrives in one sharp peak, so `CIR_PWR` is large and `STDEV_NOISE` is low relative to the peak. In NLOS, energy is spread across multiple peaks and the noise floor is elevated. Configuration constants (`CH`, `PREAM_LEN`, `BITRATE`, `PRFR`) contribute zero importance because they do not vary meaningfully across measurements in this dataset.

Balanced dataset. The equal split of 21,000 LOS and 21,000 NLOS samples avoids the class-imbalance problems that plague many real-world deployments where one condition dominates. This means accuracy is an unbiased summary metric and no oversampling (SMOTE) or class-weighted loss modifications were required.

Random Forest vs. MLP. Although the project brief references MLP as an alternative (citing Bregar and Možina

[7]), the Random Forest was selected as the primary classifier for several reasons: it produces directly interpretable feature importance scores needed for the feature-selection step; it requires no feature scaling (unlike MLP which is sensitive to input magnitude); and it achieved 91.40% accuracy without any hyperparameter tuning beyond `n_estimators=100`. The MLP would require cross-validated grid search over learning rate, hidden layer sizes, and regularisation, substantially increasing pipeline complexity.

IV. Conclusion and Future Work

A. Conclusion

This paper presented a complete data-analytics pipeline for LOS/NLOS classification and distance estimation using UWB CIR measurements. The pipeline follows the 3D process (Data Preparation, Data Mining, Data Visualisation) on a balanced empirical dataset of 42,000 samples from seven distinct indoor environments. Key contributions include: (1) a robust data-preparation stage incorporating CIR normalisation by `RXPACC` and header sanitisation; (2) a feature-importance ranking step that reduces the feature space from 14 candidates to 9 predictive signals; (3) a Random Forest Classifier achieving 91.40% LOS/NLOS test accuracy; (4) a Random Forest Regressor for Path 1 distance estimation (RMSE = 1.5027 m, R^2 = 0.5921); and (5) a novel second-path extraction procedure and corresponding Path 2 regressor (RMSE = 1.7823 m) that enriches the signal model for downstream positioning. Together these results demonstrate that ensemble tree methods can reliably exploit low-dimensional UWB diagnostic features to support accurate indoor ranging without requiring deep-learning infrastructure.

B. Future Work

Several directions can further improve the system:

- **MLP / deep learning comparison.** The current MLP classifier exhibited a high false-negative rate (1,451 samples), indicating that the model often misidentifies NLOS signals as LOS [cite: 85, 90]. Future work will involve a systematic grid search or Bayesian optimization over learning rates, hidden layer architectures, and dropout regularisation to reduce this bias and match the reliability demonstrated by the Random Forest baseline .
- **End-to-end CNN on raw CIR.** Training a 1D convolutional network directly on the 1,016-sample CIR vector could capture subtle waveform patterns that are lost in the scalar-feature projection.
- **Environment-specific fine-tuning.** The current model trains a single global classifier across all seven environments. Transfer-learning or hierarchical models could exploit environment identity to improve accuracy in each specific deployment.
- **Multi-anchor fusion.** Combining per-link LOS/NLOS decisions and corrected ranges from multiple anchors through weighted least-squares or particle-filter localisation would yield full 2D/3D position estimates and allow evaluation at the position-error level.
- **Probabilistic NLOS bias correction.** Rather than a hard LOS/NLOS binary label, a regression-based NLOS bias estimator could provide a continuous range-error correction, further reducing RMSE below the 1.5 m threshold.
- **Real-time deployment.** The trained Random Forest models are lightweight and can run on embedded processors. Packaging the pipeline as an edge-inference service on the DWM1000 host MCU would enable real-time localisation corrections without cloud connectivity.

References

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